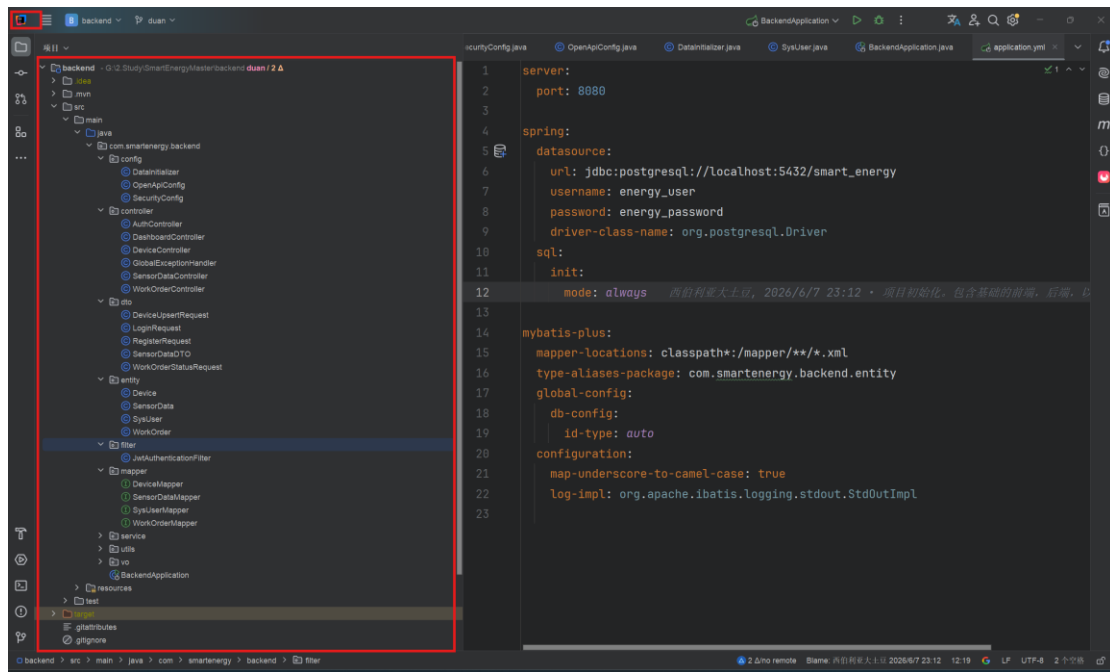
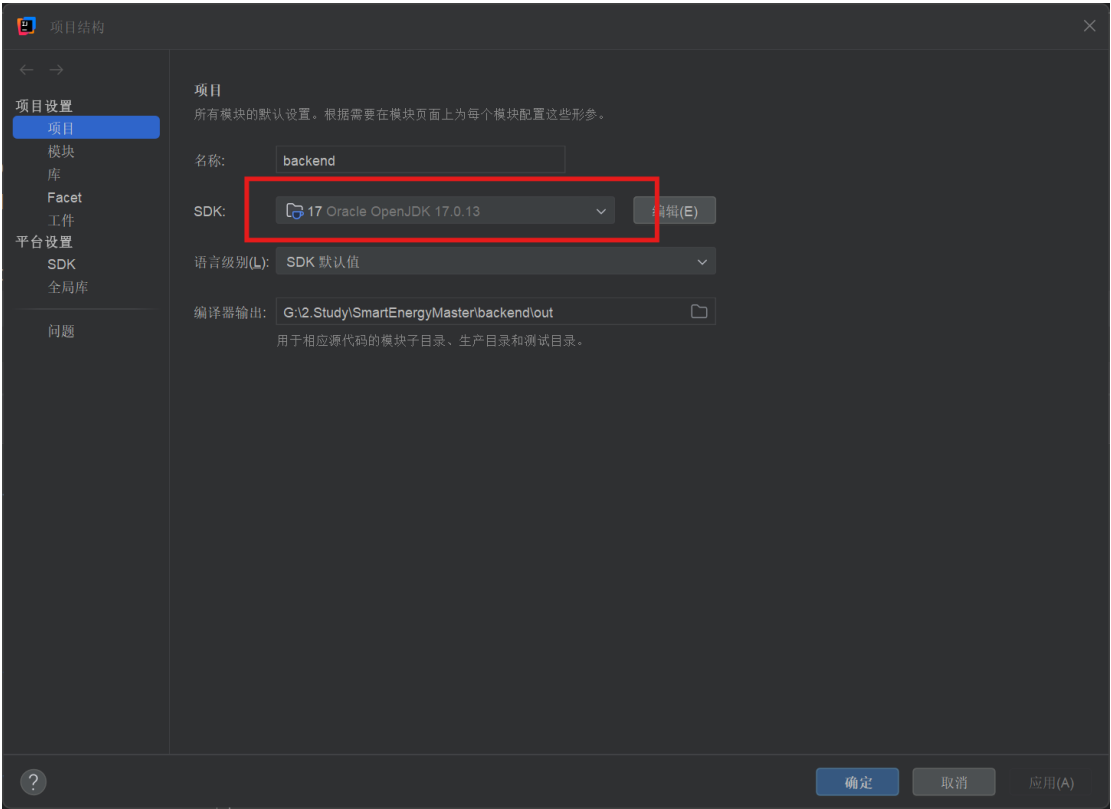


后端开发环境

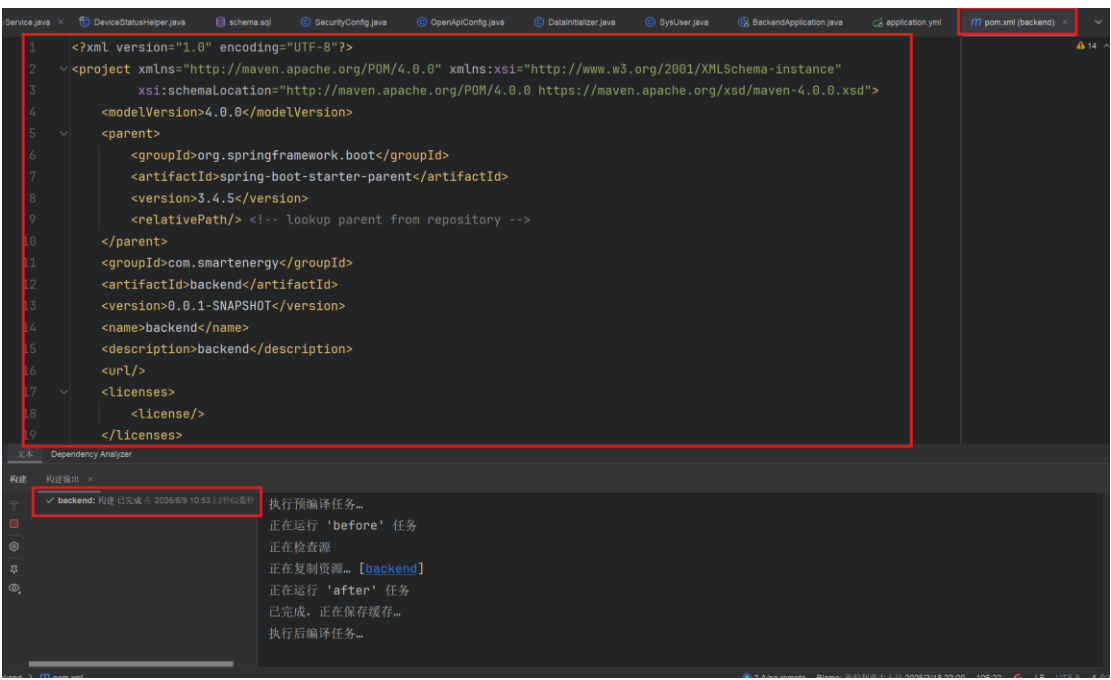
IDE 界面



JDK 版本

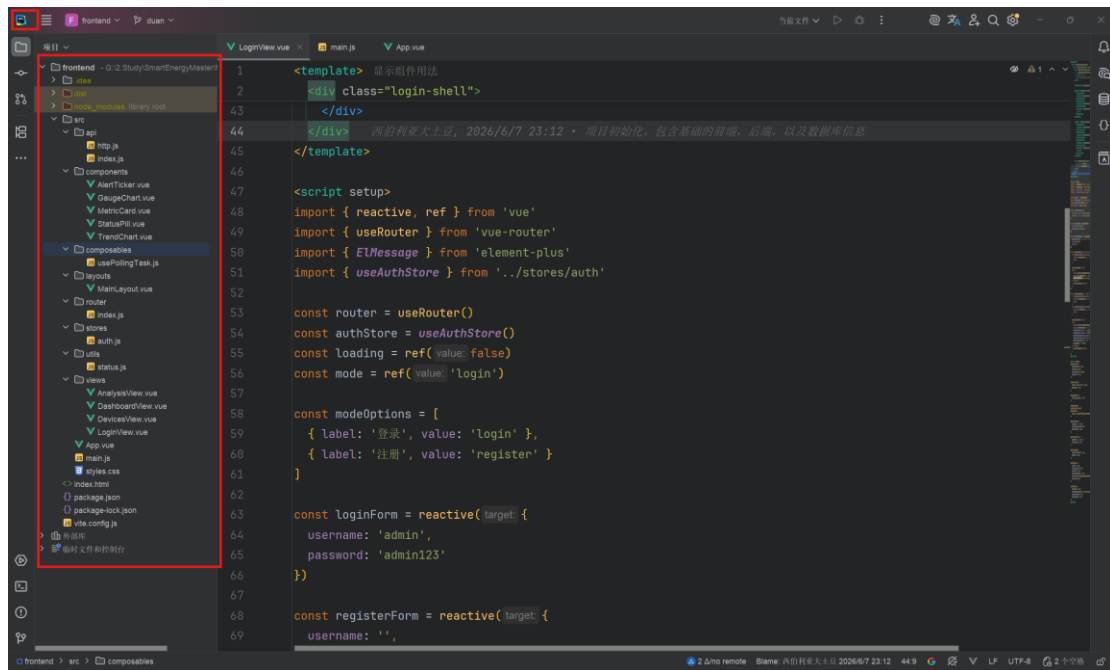


Maven 依赖

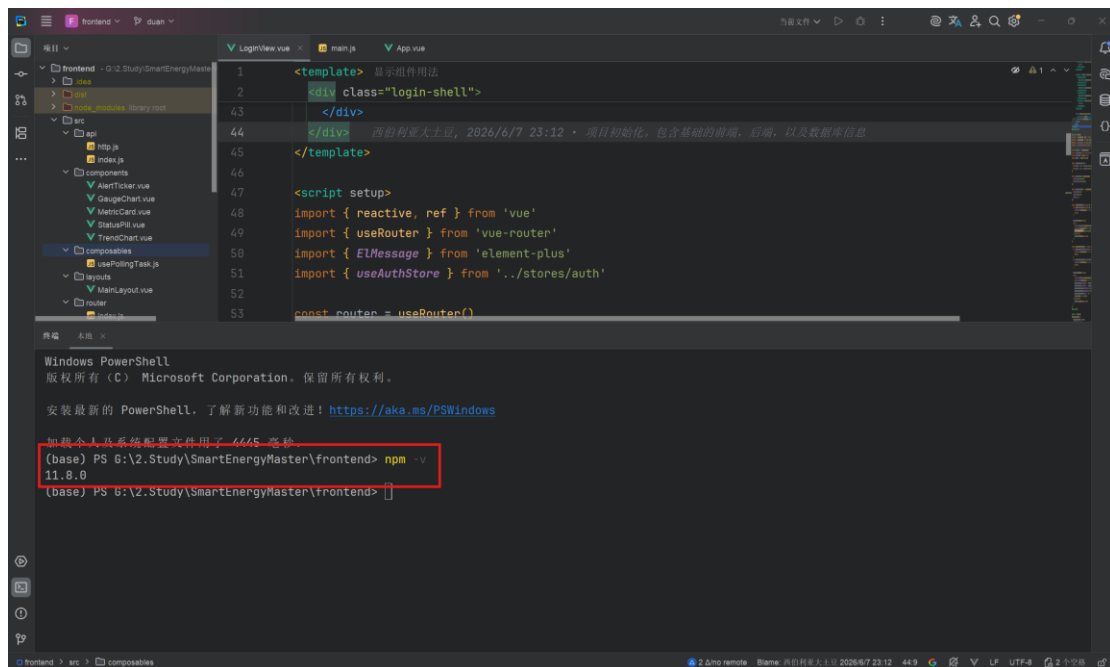


前端开发环境

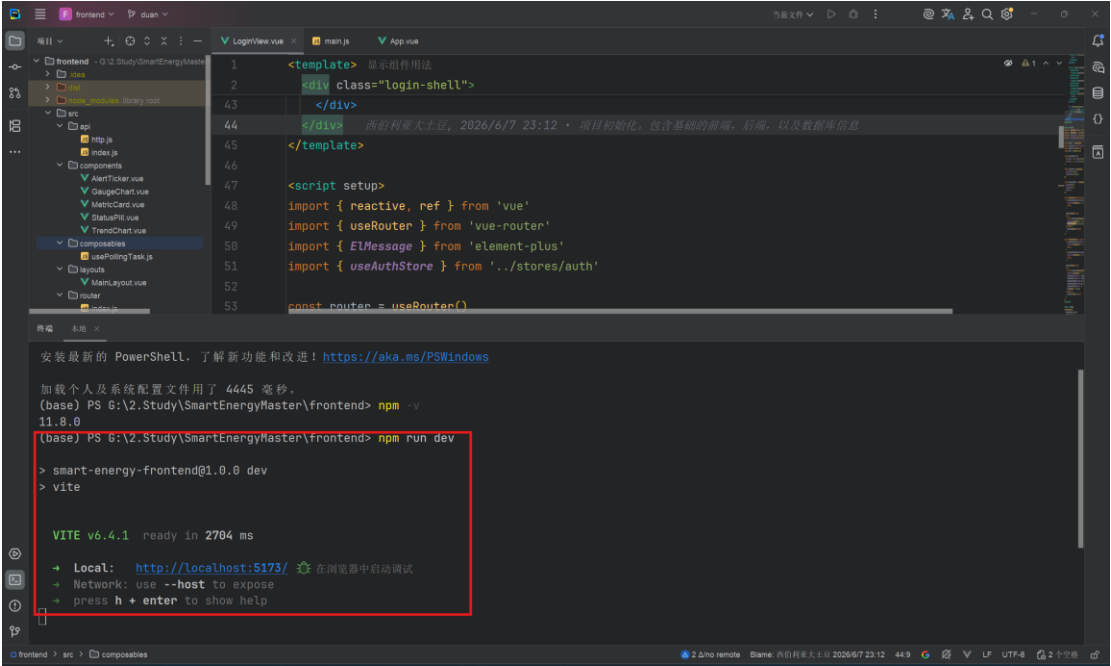
编辑器界面



Npm 环境

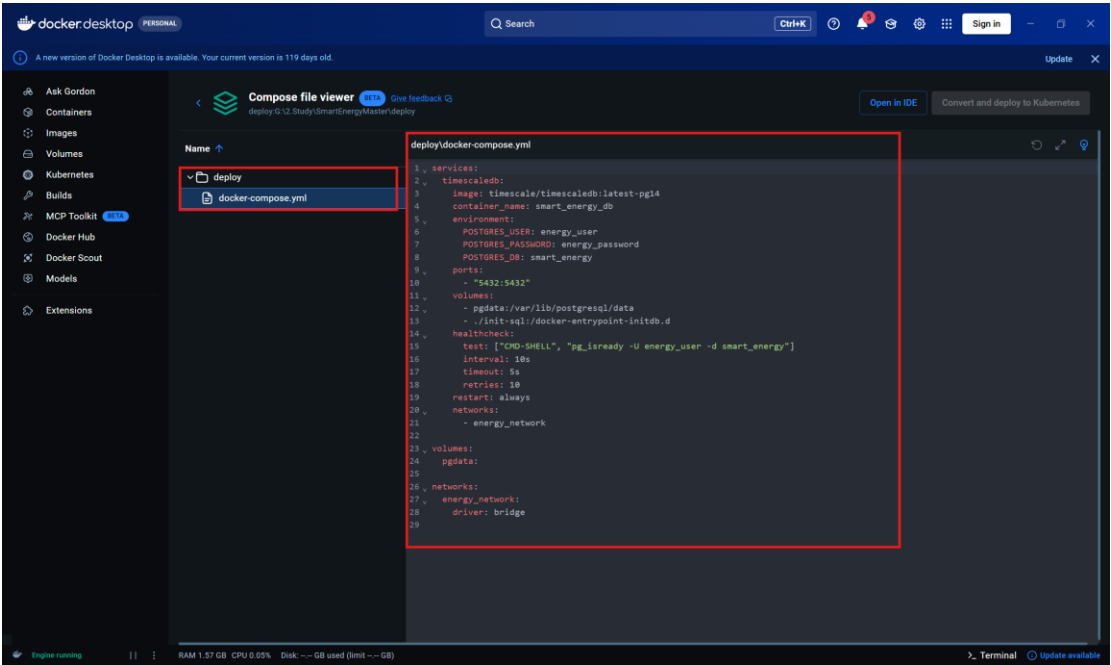


本地运行服务



数据存储环境

Docker Desktop 界面



Docker Compose 状态

```
Windows PowerShell
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安装最新的 PowerShell，了解新功能和改进！ https://aka.ms/PSWindows

加载个人及系统配置文件用了 2448 毫秒。
(base) PS G:\2.Study\SmartEnergyMaster\deploy> docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS
smart_energy_db      timescale/timescaledb:latest-pg14  "docker-entrypoint.s..."  timescaledb  36 hours ago   Up About a minute (healthy)
```

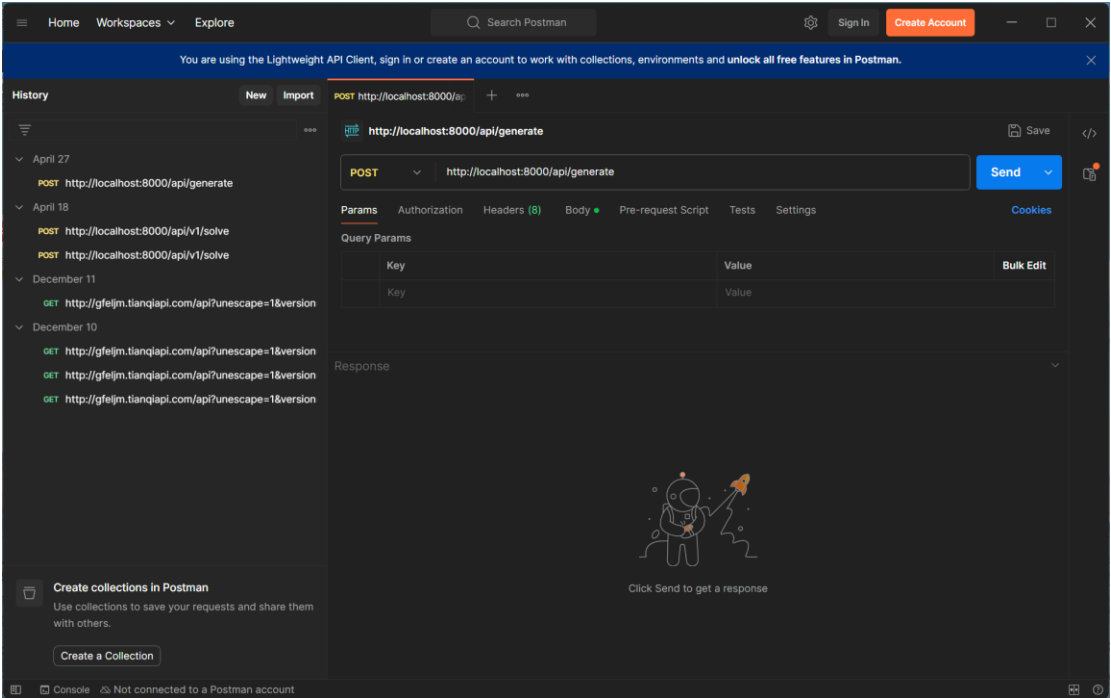
AI 与模型环境服务

Python 环境

```
# packages in environment at F:\1.Tools\Anaconda\envs\py39:
# Name                Version                Build                Channel
openmp_mutex          4.5                    2.gnu                https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
absl-py                1.0.1                 pypi_0              pypi
accelerate            0.10.1                pypi_0              pypi
aiosignal             1.3.2                 pypi_0              pypi
albucore              0.0.23               pypi_0              pypi
albumentations        1.4.24               pypi_0              pypi
annotated-doc         0.0.4                 pypi_0              pypi
annotated-types       0.7.0                 pypi_0              pypi
anyio                 4.12.1               pypi_0              pypi
asttokens             3.0.0                 pypi_0              pypi
asynctest             5.0.1                 pypi_0              pypi
attrs                 25.3.0               pyh71513ae_0        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-auth            0.8.1                 hd11252f_0          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-cal             0.8.1                 h699ea23_3          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-common          0.10.6               h2466b89_0          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-compression     0.3.0                 h699ea23_5          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-event-stream    0.5.0                 h85d8506_11         https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-http            0.9.2                 h3883f6d_4          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-io              0.15.3               hc5a9e45_6          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-mqtt            0.11.0               h2c94728_12         https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-s3              0.7.9                 h6a47413_1          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-c-sdkutils        0.2.2                 h699ea23_4          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-checksums         0.2.2                 h699ea23_4          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-crt-cpp           0.29.9               he488853_2          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
aws-sdk-cpp           1.11.489             h7d73289_0          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
blinker               1.9.0                 pyhff2d567_0        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
brotli                1.1.0                 h2466b89_2          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
brotli-bin            1.1.0                 h2466b89_2          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
brotli-python         1.1.0                 py39ha51f57c_2      https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
bzip2                 1.0.8                 h2466b89_7          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
c-ares                1.34.4               h2466b89_0          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
ca-certificates       2025.4.26             h4c7d964_0          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
cachetools            6.2.1                 pypi_0              pypi
cairo                 1.18.2               h5782b6f_1          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
certifi               2025.4.22             pypi_0              pypi
cffi                  2.0.0                 pypi_0              pypi
charset-normalizer    3.4.7                 pypi_0              pypi
click                 8.1.8                 pyh7428d3b_0        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
colorama              0.4.6                 pyh8e1d1ab_1        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
contourpy             1.3.0                 py39h2b77a98_2      https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
cryptography          46.0.6               pypi_0              pypi
cycler               0.12.1               pyh8e1d1ab_1        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
dashscope             1.25.17              pypi_0              pypi
datasets              3.4.1                 pyh8e1d1ab_0        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
dill                  0.3.8                 pyh8e1d1ab_0        https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
distro                1.9.0                 pypi_0              pypi
double-conversion     3.3.1                 he0c23c2_0          https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/conda-forge
```

互检与规范性证明

API 调试工具



Git 分支状态

```
Admin@lenovo MINGW64 /g/2.Study/SmartEnergyMaster (duan)
$ git branch -a
* duan
main
remotes/origin/Liwenzhuo
remotes/origin/dev
remotes/origin/main
```